FireTrace Sentinel — Project Report

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We have a working wildfire warning agent that replays two real 2025 LA fires and warned 100% of destroyed homes before the fire reached them; it runs on Chalk, but the AI part is not yet proven better than simple rules.

The idea in one minute

FireTrace Sentinel is an always-on agent that watches LA and Ventura for wildfires and tells people when to get ready and when to leave, before anyone has to ask.

- **It watches.** Every 15 minutes it pulls the feeds FireTrace already collects: NASA satellite fire detections, reported incidents and fire perimeters, National Weather Service Red Flag Warnings, and hourly wind, humidity and fuel dryness.
- **It judges each address.** For every subscriber it works out how far the nearest fire is, whether they are downwind, roughly how many hours until the fire could arrive, and how burnable the area is. That becomes one of four levels: none, watch, warn, evacuate.
- **It reaches out first.** When someone's level changes, it pushes an alert: a text message to a person, or a signed webhook to their personal AI agent.
- **Personal agents plug in.** Anyone's own AI assistant can connect over MCP (the standard way AI assistants call outside tools) to ask "how dangerous is it at my mom's house?" The personal agent knows the household (kids, pets, no car, oxygen machine) and turns the warning into a plan. The sentinel never stores any of that.

The pitch: official evacuation orders come late and are zone-wide. This agent warns per address, early, and explains why.

What Chalk is, and what we use

Chalk is a cloud platform for running AI agents and proving they work: it hosts your code, routes your AI model calls, and runs test suites that score the agent. We use three of its four headline pieces, and the one judges weigh most (evaluations) is where we are strongest.

Chalk piece	What it does, in plain words	How we use it	Status
Agent Evals	Runs your agent over a test set and scores every answer, so you can compare versions side by side in a dashboard	Replays Eaton and Palisades for 600 simulated households and scores each version on 5 safety metrics	Working: full runs done, 0 errors
AI Gateway (Model Gateway)	One web address that forwards your AI calls to OpenAI/Anthropic, with keys, budgets and usage tracking	The LLM version of the agent calls gpt-6.1-sol through it, using the free HackHarvard key (about 1M tokens a day)	Working: tested on 12 households
Sandboxes	Secure cloud machines that run your code 24/7	Deploy script runs the sentinel loop and the MCP server around the clock	Smoke test passed; the sentinel itself not deployed yet
Federated Queries	Pulls data from wherever it lives (databases, files) with one query	Not used: our data comes from public feeds and files	Not used

lets cache this data in the sources especially the historical stuff

Are we using it well? Evals are a genuine fit, not decoration: the whole question "does the agent warn people in time?" is answered inside Chalk. The gateway is used for real model calls. Sandboxes are only half-done until the sentinel actually runs there.

Is it working?

Yes, the core works end to end on real data; the gaps are deployment and the AI version's results, not the foundation.

Piece	What it is	Status
Threat scoring	Distance, downwind, hours-to-arrival, wind, fuel, Red Flag per address	Working, tested on live and historical data
Replay of past fires	Re-lives Eaton and Palisades hour by hour, seeing only what was known at each hour	Working
Rules version	Fixed thresholds decide the level	Working, fully evaluated
LLM version	gpt-6.1-sol decides the level and writes the message	Working; only 12 households tested (token budget)
Alerts out	Signed webhooks to agents, SMS to people	Webhooks tested and verified; SMS needs a Twilio account
MCP server	Tools personal agents call (get_threat, subscribe, replay_threat…)	Working, tested locally; not yet tried with a real personal agent such as Claude Desktop
Live 24/7 loop	Refreshes feeds every 15 min and alerts subscribers	Works on one machine; not yet deployed to a Chalk sandbox
Chalk evaluation suite	The replay test run on Chalk with 5 scorers	Working: 600 households per version, 0 errors
Message-quality judge	An AI grader that scores alert wording	Written, never run
Web app and landing page	Map plus a page presenting the replay evaluation	Built in a separate session and merged; not reviewed in this report

Code is pushed to github.com/Berny-ft/HackHarvard2026 (agent in `agent/`). The latest token-saving changes to the LLM version are not committed yet.

Results so far

Every version warned 100% of destroyed homes before the fire reached them, about 9 hours ahead; the differences are in who got told to evacuate in time.

How the test works. We replay the Eaton and Palisades fires (January 2025) hour by hour. The agent only sees what was known at that hour. For each home, the moment the fire "arrived" is the first NASA satellite fire detection within 750 m, so the answer key comes from real observations, not from our own model. The test set is 600 locations: 200 homes CAL FIRE recorded as destroyed, 200 undamaged homes near the fire, and 200 places 2–20 km outside the burn area where the fire never went.

Version	Warned in time	Told to evacuate in time	Typical evacuate head start	Needless evacuations	False warnings far away
Distance rings only (baseline)	100%	92.5%	4.4 h	4%	6%
Wind-aware rules	100%	82.5%	4.4 h	4%	6%
LLM (12 households only)	100%	75% (rules: 75% on the same 12)	4.9 h	0%	0%

What this honestly says:

- **The baseline wins on evacuations.** Simple distance rings beat our wind model because they trust every fire report immediately. Our rules wait for a satellite to confirm a fire before saying "evacuate", which costs about 10 points, because the first satellite pass came 2–3 hours after ignition.
- **That caution is deliberate.** Right now 48 "active" incidents are listed around LA, most of them tiny. Trusting every report would send evacuation alerts all over the city. Deciding which reports to trust is exactly the judgment an LLM should be good at.
- **The LLM writes much better messages** (specific distance, direction, gusts, embers) but in the small test it was more cautious early, halving the warning head start (4.9 h vs 10.9 h). Twelve households is too few to conclude anything.
- **The wind physics adds little on these two fires.** Satellites only see the fire every few hours, so the agent rarely knows where the front is. Tuned on Eaton and checked on Palisades, the fitted model ends up close to the distance rule.

Limits: only two fires; arrival times are only as precise as satellite passes; hourly weather stands in for a forecast, which flatters every version a little.

Is it a good idea?

Yes for this challenge: it hits all three judging criteria with real data and a real test, and its weak spot (the AI not yet beating rules) is fixable and can itself be presented as an honest finding.

Judging criterion	Our case	Strength
Impact	CAL FIRE records 9,419 structures destroyed in Eaton and 6,845 in Palisades (Jan 2025); about 30 people died (approximate, from memory); earlier, per-address warnings are an obvious good	Strong
Use of data	Live satellite, incident, weather and warning feeds; 25 years of fire history; real CAL FIRE damage records as the answer key	Strong
Rigour of evaluation	Replays that only see past data, an answer key independent of our model, a simple baseline to beat, one fire for tuning and the other for checking, results run in Chalk Evals	Strong, and rare at a hackathon
Does the AI add value?	Not shown yet: rules match or beat it so far	Weak until the LLM run is done

The risk to the pitch is a judge asking "why do you need an LLM at all?" Two good answers we can back with data once tested: the LLM can decide which unconfirmed reports to trust (the 10-point gap above), and it writes clearer, more specific warnings (the message-quality judge would measure that).

What to do next

The single most valuable step is a fair LLM-versus-rules comparison on about 30 households; everything else is polish or demo.

1. **Tune the LLM prompt** so that under a severe Red Flag Warning an unconfirmed report within 8 km means at least "warn", and fix confusing "upwind" wording. Cost: none.

2. **Run LLM vs rules on 30 households in Chalk Evals, with the message-quality judge.** About 100k tokens, a tenth of the daily budget. This answers "does the AI add value?"
3. **Deploy the sentinel to a Chalk sandbox** so it genuinely runs 24/7. This completes the Sandboxes piece.
4. **Connect a real personal agent** (Claude Desktop) to the MCP server and record it answering "should my family leave?" with household context. This is the demo moment.
5. **Commit and push** the token-saving changes.
6. **Optional:** SMS via Twilio; a third fire to test on.